

TASK 5 REPORT – CAPSTONE PROJECT & INCIDENT RESPONSE

Title:

Security Assessment and Incident Response Simulation

Objective:

To apply cybersecurity skills in a capstone project by performing assessment, documenting findings, and simulating an incident response process.

Project Selected:

Web Application Security Assessment on DVWA Test Environment

Tools Used:

- Kali Linux
- DVWA
- Nmap
- Burp Suite
- Browser Tools
- Firewall / Logs

Methodology:

1. Defined scope and objectives.
2. Performed reconnaissance and testing.
3. Collected screenshots and evidence.
4. Suggested remediation steps.
5. Simulated incident response lifecycle.

Activities Performed:

1. Planning

Defined target, tools, and testing timeline.

2. Security Assessment

Performed scanning and web vulnerability testing.

3. Findings Documentation

Recorded issues with screenshots and observations.

4. Incident Response Simulation

- Detection of issue
- Containment

- Eradication
- Recovery
- Lessons learned

5. Final Documentation

Prepared final report and evidence.

Findings:

- Weak input validation risks identified.
- Misconfiguration and exposure points reviewed.
- Security awareness importance highlighted.

Mitigation:

- Patch systems regularly.
- Use secure coding practices.
- Enable security headers.
- Monitor logs and alerts.
- Train users against phishing/social engineering.

Conclusion:

The capstone task combined technical testing and incident response planning, demonstrating practical cybersecurity workflow skills.

Result:

A complete mini security assessment with incident response documentation was successfully completed.